Team ID : 973727

Arete Services Private Limited

AN INTERNSHIP REPORT

Submitted by:

Patel Niket

220280116092

In partial fulfillment of the award of the degree of

BACHELOR OF ENGINEERING

in

Information Technology

L.D. COLLEGE OF ENGINEERING

NAVRANGPURA, AHMEDABAD

Gujarat Technological University, Ahmedabad

[April, 2026]

Team ID : 973727

L.D. COLLEGE OF ENGINEERING

NAVARANGPURA,AHEMDABAD

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **Ticket Raising System** has been carried out by **Patel Niketkumar Shaileshbhai (Enrollment No. 220280116092)** under my guidance in partial fulfillment for the degree of Bachelor of Engineering in Information Technology, 8th Semester of Gujarat Technological University, Ahmadabad during the academic year 2026.

Prof.Bakul Panchal
Internal Guide

Dr.H.M.Diwanji
Head of the Department

arete
values create value

Date: 31st March 2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Mr. Niket Patel has successfully completed his Internship in IT from 01st January 2026 to 31st March 2026.

During this period, he was found hard working and sincere.

We wish him all success in his future endeavor.

Yours Sincerely,

For Arete Group of Companies

Tanaya

Authorized Signatory

ARETE SERVICES PRIVATE LIMITED ★ VAPI

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L.D. COLLEGE OF ENGINEERING
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DECLARATION

We hereby declare that the Internship report submitted along with the Project entitled **Ticket Raising System** submitted in partial fulfillment for the degree of Bachelor of Engineering in to Gujarat Technological University, Ahmedabad, is a bona-fide record of original project work carried out by me at **Arete Services Private Limited** under the supervision of Internal Guide Prof. BAKUL PANCHAL and External Guide KRUPAL PATE**L** and that no part of this report has been directly copied from any student's reports or taken from any other source, without providing due reference.

Patel Niket

Name of the Student   Sign of Student

ACKNOWLEDGEMENT

I would like to expand my heartiest much appreciated to Mr. KRUPAL PATEL (Arete Services Private Limited) for giving me an opportunity for an internship at Arete Services Private Limited and supporting me in fact amid the period of internship and for supporting us amid the internship period. They guided me all the time and persuaded me inside their active plan.

Whole heartedly, I would like to thank Prof. BAKUL PANCHAL (Information Technology) for giving me the nonstop back and direction to seek after my last semester internship at “Arete Services Private Limited” whereas picking up down to earth working encounter within the industry. In expansion, all the resources from our division given us a favorable environment and fundamental direction, without them I would not have accomplished my objective. They have continuously been accessible for any help and were continuously an awesome source of motivation for me.

Patel Niket
220280116092

ABSTRACT

Title: Ticket Raising System Platform

Objective: The Ticket Raising System is a full-stack web application designed to manage and streamline the process of raising, assigning, and resolving support tickets within an organization. It provides a centralized digital platform where users can submit issues, and administrators or developers can track, update, and manage their status efficiently. The system improves communication between users and support teams, ensures transparency in ticket handling, and enhances overall workflow management through structured digital tracking.

Scope: Users can register and log in with different roles—Admin, Developer, and User—each with specific functionalities. Users can raise tickets by submitting issues and track their status. Admin can assign, reassign, and manage tickets, monitor overall workflow, and control user access. Developers can view assigned tickets, update status, add task details, and respond through comments. The system maintains proper role-based access control to ensure secure and structured operations. Additional features include ticket reassignment with reason tracking, task management, comment notifications, and status updates such as Open, In Progress, Resolved, and Closed. Users can also manage their profiles for a personalized experience. The application is designed to be responsive and perform smoothly across different devices while using AJAX-based operations for better user interaction without page reloads.

Technology Stack:

- **Frontend:** HTML, CSS, Bootstrap, JavaScript, jQuery, AJAX
- **Backend:** PHP (CodeIgniter Framework)
- **Database:** MySQL
- **Authentication:** Session-based authentication (Role-based access control)
- **API Communication:** AJAX-based asynchronous requests

Key Features:

1. **User Authentication:** Registration, login, logout, and profile management with role-based access control (Admin, Developer, User).
2. **Ticket Management:** Users can raise, edit, and track tickets, while Admin can manage and monitor

   all tickets in the system.
3. **Comment & Notification System:** Users and developers can communicate through ticket comments, and notifications are generated for ticket-related updates.
4. **Task Management:** Developers can create and update tasks related to tickets, helping in structured issue resolution.
5. **Status & Assignment Management:** Tickets are categorized by status (Open, In Progress, Resolved, Closed), and Admin can assign or reassign tickets to developers with proper tracking.

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CHAPTER 1: OVERVIEW OF THE COMPANY

1.1 HISTORY

At **Arete Services Private Limited**, we believe in the transformative power of technology. As a leading IT consulting firm based in Vadodara, Gujarat, India, the company is dedicated to helping businesses navigate the ever-evolving digital landscape. The journey of the organization began in 2016 with a clear vision to deliver innovative and practical consultancy solutions that drive business growth, improve operational efficiency, and support long-term success.

In today's fast-paced digital world, businesses require more than just technical solutions—they need strategic guidance tailored to their specific needs. Arete Services focuses on understanding each client's challenges and objectives before recommending solutions. With a strong commitment to personalized service and technological expertise, the company provides strategies that align with business goals and ensure sustainable development.

1.2 SCOPE OF WORK

At **Arete Services Private Limited**, the core services include Business Consultancy Services and Corporate Training, delivered through both online and offline modes. The company offers consultancy in areas such as Software Consultancy, Website Consultancy, Mobile Application Consultancy, CRM Consultancy, and ERP Consultancy. The primary objective of these services is to guide businesses in selecting the most suitable tools and technologies for their growth and operational needs.

The consultancy services also include one-to-one business growth consultation, lead generation strategies, selection of appropriate social media platforms for brand expansion, and brand-building strategies.
If you are not familer about the how Consultant can help you, then here is few list of points we can help you with. 1 to 1 Business Growth Consultation, Lead Generation, Selection of Social Media Platform for your Business Growth, Brand Building and many more.

In addition to consultancy, the company provides Corporate Training programs for college students, freshers, faculty members (FDP programs), and industry professionals. Training areas include Computer Fundamentals, Web Development (HTML, CSS, JavaScript, ReactJS, ASP.NET, Java, JSP), Mobile Application Development (Android, iOS, Flutter), SEO & Digital Marketing, Cloud Computing, and other technical domains.

1.3 ORGANIZATIONAL CHART

Fig1.1 Organization chart

1.4 CAPACITY OF PLANT

Arete Services Private Limited operates with a team of skilled professionals and industry experts, establishing itself as a trusted IT consultancy and training organization. The company manages projects from initial concept understanding to documentation preparation and coordination with technical teams to ensure successful project execution. Its operational capacity enables it to handle consultancy, planning, and technical guidance across multiple domains efficiently.

The organization follows a collaborative and integrity-driven approach, working closely with clients to understand their operational structure and business objectives. By analyzing business processes in detail, the company identifies opportunities for growth, optimization, and automation. It specializes in Sales and Marketing Planning, Business Growth Strategy, IT Consultancy, and Software Consultancy services such as CRM, ERP, Inventory Management Systems, and other enterprise solutions. The company provides structured roadmaps and strategic guidance to help businesses achieve their long-term goals effectively.

CHAPTER 2: OVERVIEW OF DIFFERENT DEPARTMENT OF THE ORGANIZATION

2.1 DIFFERENT DEPARTMENTS

Arete Services Private Limited consists of multiple specialized departments, each focused on delivering specific IT and consultancy services to meet diverse business requirements. The major departments of the organization are as follows:

1. **Website Design and Development Services**: This department focuses on creating modern, responsive, and user-friendly websites tailored to business needs. The websites are optimized for performance, security, and search engines to ensure smooth user experience across all devices.

2. **CRM & ERP Software Development**:

i. Streamline Business Processes
ii. Provide Custom Software Solutions
iii. Enable Business Growth
iv. Ensure User-Friendly Interfaces

3. **Google Business Registration & Improvement**: This unit assists businesses in registering and optimizing their Google Business profiles to improve local search visibility and generate quality leads.

4. **Domain and Hosting Service:**

i. Secure Your Domain Name
ii. Reliable Hosting
iii. 24/7 Technical Support

5. **Business Email (Free and Paid Both Available)**:

i. Professional Communication Solutions
ii. Secure and Reliable Email Hosting
iii. Customized Email Services
iv. Enhancing Business Credibility

6. **Shopify Website Development and Maintenance:**

i. Online Store Development
ii. Ongoing Maintenance Support
iii. Custom E-commerce Solutions
iv. Easy Store Management

7. **Social Media Promotion Services:**

i. Increase Brand Visibility
ii. Customer Engagement
iii. Targeted Advertising Campaigns
iv. Performance Tracking and Analysis

8. **UI/UX Design Services:** This department transforms business ideas into visually appealing and user-centric designs. UI/UX services are provided independently or integrated within mobile and web application development projects.

- Consultancy Services Department

1. Market Research Consultant
2. Business Consultant
3. Website Development Consultant
4. Marketing Consultant
5. Digital Marketing Consultant
6. CRM Consultant
7. ERP Consultant
8. Education Consultant
9. Software Consultant
10. IT Consultant
11. Technology Consultant

2.2 LIST THE TECHNICAL SPECIFICATIONS OF MAJOR EQUIPMENT USED IN EACH DEPARTMENT

- Programming Language : PHP, Java, C, C++, VB6, ASP, VB.NET, FoxPro, HTML, Dart, Android (Java/Kotlin), Excel Macro Programming
- Programming Framework & Tools : Laravel , Codeigniter
- Database Technologies: MySQL, Microsoft SQL Server, Oracle, SQLite, MS Access, MS Excel
- Operating Systems: MS-DOS (including commands and batch files), Windows 9x, Windows XP, Windows 7, Linux, Android
- Office & Documentation Tools: Microsoft Office, Open Office, Google Drive
- Pre-Sales & Communication Tools: MS Outlook
- File Transfer Tools : Google Drive, Dropbox, 4shred, One Drive
- Mockup & Design Tools: Microsoft PowerPoint, Figma
- Conference & Remote Access Tools: TeamViewer, Skype, Remote Admin, Remote Desktop, Zoom, Google Meet
- Messenger & Collaboration Tools: Skype, Google Hangout
- File Transfer & Cloud Storage Tools: Google Drive, Dropbox, 4shared, OneDrive Core FTP, WS FTP, FileZilla
- Website Management Panels: cPanel, Website Panel (Cron Job, Scheduled Task, File Manager, FTP, Database Management)
  Plesk Panel, WHM Panel
- Data Collection Tools: Google Forms, JotForm
- Virtual Environment & Testing Tools: VMware, Microsoft Virtual PC
- Screen Capture & Recording Tools: Snipping Tool, Easy Capture, Windows Media Encoder
- CRM & ERP Software Tools: Microsoft dynamics 365(F & O), Tally
- Business Email Platforms: cPanel Email,Google Workspace, Microsoft365
- Blogging Platforms: WordPress, Blogger, Wix
- **AI & Automation Tools:** ChatGPT, Google Gemini
- **Web Browsers:** Google Chrome, Mozilla Firefox, Opera,Microsoft Edge

2.3 PREPARE SCHEMATIC LAYOUT OF END PROJECT

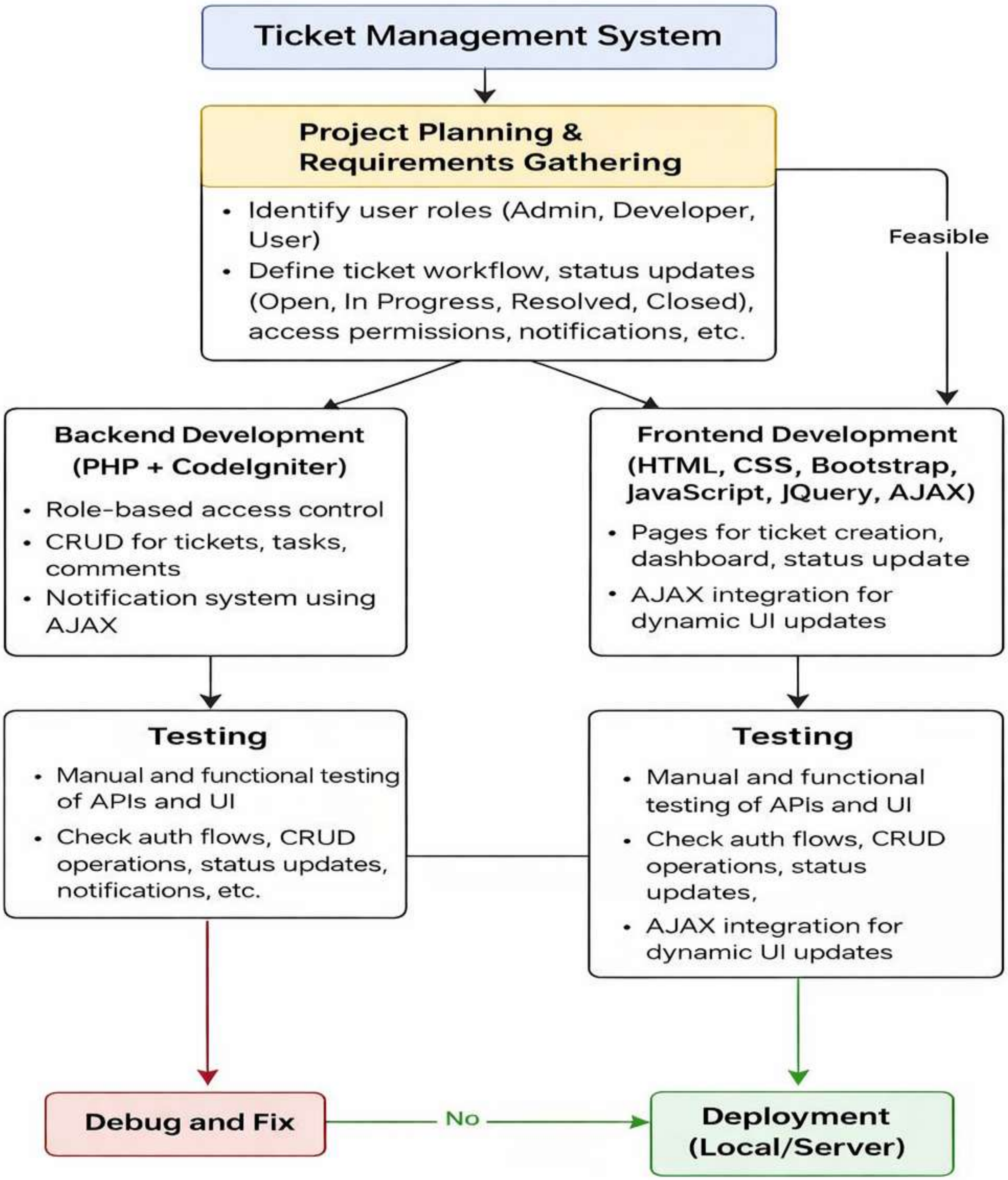

Fig 2.1 Schematic Layout of Operation for Development

2.4 EXPLAIN IN DETAILS ABOUT EACH STAGE OF PRODUCTION

ANALYSIS

- During this stage, all project requirements for the Ticket Raising System were gathered and analyzed in detail. The team studied both functional and non-functional requirements for a system that allows users to raise tickets, developers to resolve them, and admins to manage the overall workflow. A role-based access model was finalized to distinguish Admin, Developer, and User functionalities. The system's core goals included user authentication, ticket creation and management, task handling, status updates (Open, In Progress, Resolved, Closed), comment notifications, and reassignment tracking. A high-level architecture diagram was created to represent modules such as authentication, ticket management, task management, and notification handling.

PLAN

- In this stage, time and resource estimation was performed for each development activity. Features such as user registration/login, ticket CRUD operations, role-based access control, task management, comment system, and AJAX-based updates were broken down to estimate effort. The technology stack (PHP with CodeIgniter, MySQL, HTML, CSS, Bootstrap, JavaScript, jQuery, AJAX) was selected to support a structured and maintainable web application. A development timeline was prepared including milestones such as database design, backend development, frontend integration, testing, debugging, and final deployment. The scope was clearly defined to ensure project completion within the internship duration.

DESIGN

- The system design follows a client-server architecture. The frontend is developed using HTML, CSS, Bootstrap, JavaScript, and jQuery, while the backend is built using PHP with the CodeIgniter framework. MySQL is used as the relational database. The database schema was designed with tables such as users, tickets, tasks, ticket_comments, and ticket_messages. Role-based access control was implemented to restrict functionalities based on user roles. Wireframes were prepared for login, dashboard, ticket creation form, admin panel, and developer dashboard. UI flow and AJAX-based dynamic updates were planned to ensure a smooth and responsive user experience without page reloads.

DEVELOP

The development phase involved implementing both frontend and backend modules:

- PHP (CodeIgniter) was used for backend logic, including authentication, session management, ticket CRUD operations, task updates, reassignment handling, and notification processing.

- MySQL database was configured to manage structured data with proper relationships between users, tickets, and tasks.

- AJAX and jQuery were implemented to handle asynchronous operations such as updating ticket status, adding comments, reassigning tickets, and loading notifications without refreshing the page.

- Bootstrap was used to design responsive dashboards, forms, modals, and tables for Admin, Developer, and User Panels.

TESTING

- Testing was performed to ensure the system met functional and usability requirements. Backend functionalities such as authentication, ticket creation, update, reassignment, and status management were tested thoroughly. Manual testing was conducted to verify role-based restrictions and proper session handling. AJAX requests were tested to confirm that ticket updates, comments, and notifications worked correctly without page reload. Database integrity was verified to ensure proper data storage and retrieval. Bugs related to validation errors, session handling, and UI responsiveness were identified and resolved.

LAUNCH

- After successful testing and debugging, the system was prepared for deployment. The application can be hosted on a local server (XAMPP) or deployed to a live hosting server supporting PHP and MySQL. The database was optimized, and final validations were performed to ensure secure login, correct role permissions, and proper ticket workflow execution. Once verified, the system became fully functional and ready for use within the organization to manage and track support tickets efficiently.

CHAPTER 3: INTRODUCTION TO INTERNSHIP

3.1 INTERNSHIP SUMMARY

During my internship at **Arete Services Private Limited**, I worked as a PHP Developer and contributed to the development of a full-stack web application titled **Ticket Raising System (TRS)**. My responsibilities included designing and developing user interfaces using HTML, CSS, Bootstrap, JavaScript, and jQuery, as well as implementing backend logic using PHP with the CodeIgniter framework. I managed database operations using MySQL and performed asynchronous updates using AJAX to enhance user experience.

Throughout the development process, I utilized tools such as VS Code, XAMPP, phpMyAdmin, GitHub, and Postman for API testing and debugging.

This project provided me with real-time experience in developing a web-based ticket management platform from scratch. I implemented features such as role-based access control (Admin, Developer, User), user authentication, ticket creation and management, task handling, reassignment with reason tracking, comment system, and AJAX-based notifications. The final product is intended for organizational use and follows structured workflow management practices.

3.2 PURPOSE

The purpose of the internship was to strengthen my practical knowledge of full-stack web development using PHP and related technologies, and to understand the complete lifecycle of designing, developing, testing, and deploying a web-based application.
Through this experience, I aimed to enhance my backend development skills, database design knowledge, frontend integration techniques, and understanding of role-based access control. Additionally, the internship provided exposure to real-world problem-solving, debugging, system optimization, and collaborative development practices.

3.3 OBJECTIVE

- Gain hands-on experience in developing and deploying a full-stack web application using PHP (CodeIgniter), MySQL, HTML, CSS, Bootstrap, JavaScript, and AJAX.
- Learn how to design and implement secure authentication and session management systems.
- Understand role-based access control and its implementation in a multi-user environment (Admin, Developer, User).
- Develop CRUD operations for tickets, tasks, and comments with proper validation and database relationships.
- Improve debugging, performance optimization, and security practices in web application development.
- Gain experience using development tools such as VS Code, XAMPP, phpMyAdmin, Postman, and GitHub for efficient project management.

3.4 SCOPE

- Design and develop the infrastructure of the Ticket Raising System with distinct user roles (Admin, Developer, User).
- Implement core functionalities including user registration, login, ticket creation, ticket assignment, status updates, task management, and comments.
- Develop a responsive user interface using Bootstrap to ensure compatibility across devices.
- Integrate backend logic with frontend components using AJAX to enable dynamic updates without page reload.
- Manage and optimize structured data storage using MySQL, ensuring proper relationships between users, tickets, and tasks.
- Implement secure authentication and authorization mechanisms to maintain data integrity.
- Deploy the system on a local server environment (XAMPP) and prepare it for live server deployment.

3.5 TECHNICAL AND LITERATURE REVIEW

The Ticket Raising System project was developed using structured web technologies to ensure scalability, maintainability, and performance.

3.5.1 PHP (CodeIgniter Framework)

PHP is a server-side scripting language widely used for web development. The CodeIgniter framework was used to implement the MVC (Model-View-Controller) architecture, which separates business logic, user interface, and database operations. This structure improved maintainability and code organization.

3.5.2MySQL

MySQL is a relational database management system used to store structured data. It was used to manage tables such as users, tickets, tasks, comments, and notifications. Proper indexing and relational mapping ensured efficient data retrieval and integrity.

3.5.3HTML, CSS, and Bootstrap

HTML and CSS were used to design the basic structure and styling of the web pages. Bootstrap framework was used to create responsive layouts, dashboards, forms, and modals, ensuring cross-device compatibility.

3.5.4JavaScript, jQuery, and AJAX

JavaScript and jQuery were used to handle client-side logic and DOM manipulation. AJAX was implemented to perform asynchronous operations such as ticket updates, comments, reassignment, and notifications without refreshing the page, enhancing user experience.

3.5.5Development and Testing Tools

Tools such as VS Code (code editor), XAMPP (local server), phpMyAdmin (database management), Postman (API testing), and GitHub (version control) were used during development and testing phases.

3.5.6Additional Features Implemented

- Ticket Assignment & Reassignment System: Admin can assign or reassign tickets to developers with reason tracking.
- Task Management System: Developers can create and update tasks related to tickets.
- Comment & Notification System: Users and developers can communicate via comments with real-time notification updates using AJAX.
- Role-Based Access Control: Admin, Developer, and User have different permissions and system access levels.
- Profile Management: Users can update personal details and manage their account securely.
- Status Tracking System: Tickets are categorized as Open, In Progress, Resolved, and Closed for structured workflow monitoring.

3.6 INTERNSHIP PLANNING

3.6.1 Development Approach and Justification

Software project planning starts before the technical development begins. It helps define system requirements, project scope, technology selection, module division, and time estimation to ensure smooth execution.

The first image shows the major steps of project planning followed in the Ticket Raising System. The second image represents the Software Development Life Cycle (SDLC) model used during the system development process.

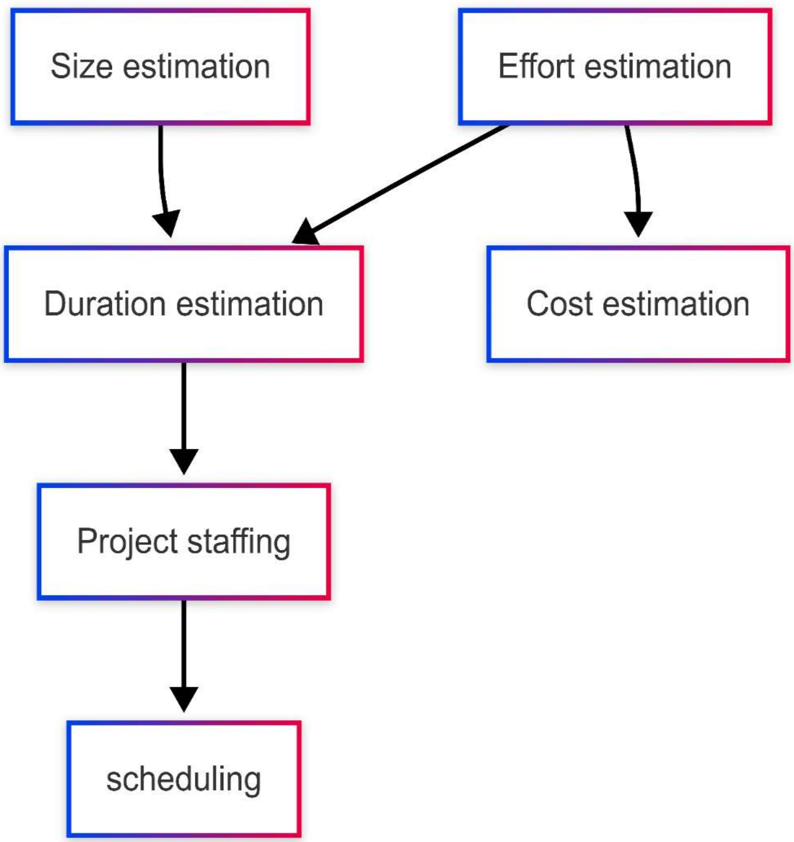

Fig 3.1 Project Scheduling

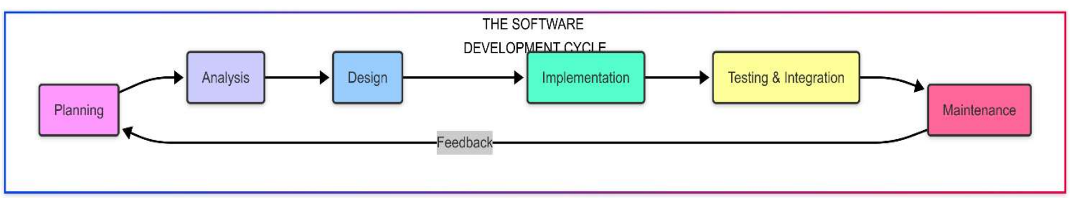

Fig 3.2 SDLC

3.6.2 Effort, Time and Cost Estimation

Cost required to develop project=effort*RS/month KLOC= Kilo Line of Code

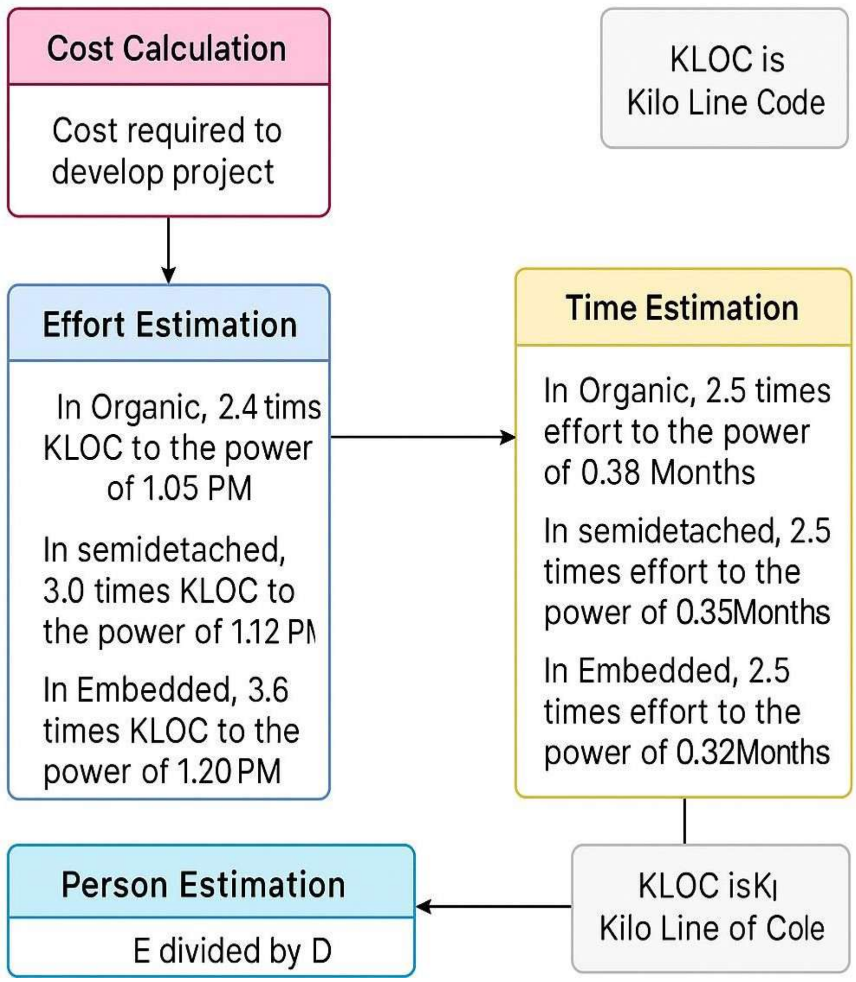

FIG 3.3 COCOMO MODEL

3.6.3 Role and Responsibilities

- Investigation
- Requirement Analysis
- Design
- Coding
- Testing

3.6.4 Group Dependencies

- Individual Dependencies

3.7 INTERNSHIP SCHEDULING (GANTT CHART)

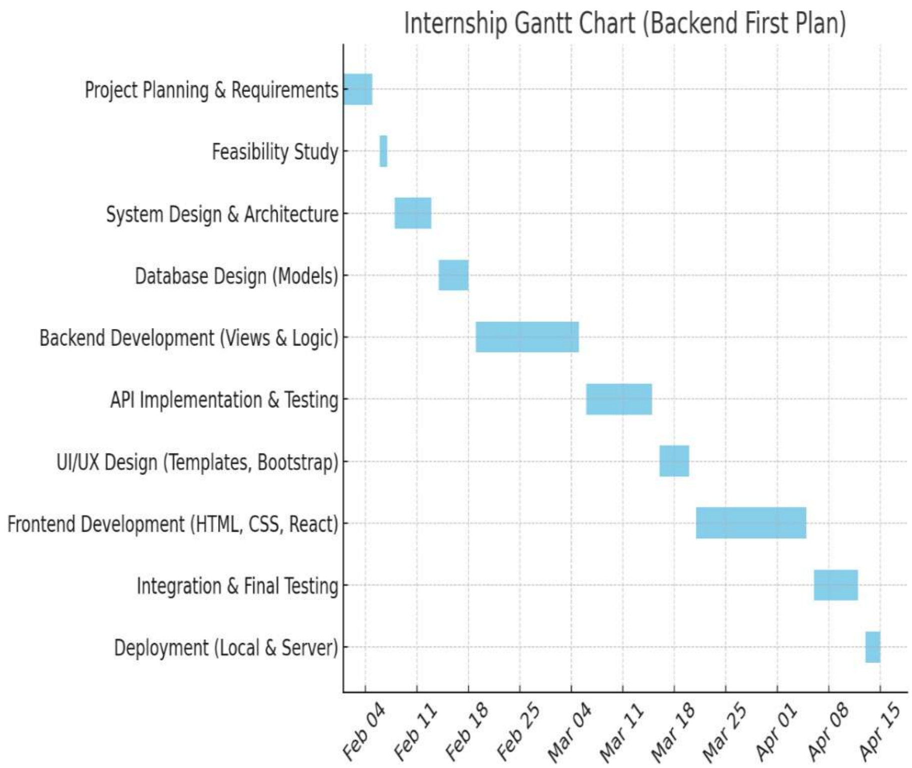

Fig 3.4 Gantt Chart

CHAPTER 4: SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

- In many organizations, issue tracking and complaint handling are managed manually through emails, phone calls, or informal communication methods. There is no centralized system to track the status of issues, assign responsibilities, or monitor resolution progress.
- The absence of a structured Ticket Raising System leads to confusion in task allocation, delayed responses, lack of accountability, and difficulty in tracking historical records of issues.
- Without a proper system, administrators cannot monitor developer performance or track the number of pending, resolved, or closed tickets efficiently.
- Communication between users and developers is often unorganized, resulting in repeated queries and mismanagement of issues.

4.2 PROBLEM AND WEAKNESS OF CURRENT SYSTEM

- **Lack of Centralized Tracking**
  Issues are not stored in a structured database, making it difficult to track ticket history and resolution status.
- **No Role-Based Access Control**
  There is no clear separation of responsibilities between Admin, Developer, and User, leading to confusion and security risks.
- **Manual Status Updates**
  Ticket progress is not updated systematically, resulting in delayed resolution and lack of transparency.
- **Poor Communication Mechanism**
  There is no proper comment or notification system to ensure smooth communication between users and developers.
- **Limited Monitoring Capability**
  Administrators cannot easily monitor pending tickets, reassigned tasks, or overall system workflow.
- **Data Security Risks**
  Manual systems or basic platforms may not provide proper authentication, session management, and secure data handling.

4.3 REQUIREMENTS OF NEW SYSTEM

- **Multi-User Authentication**
  Secure login and registration system with role-based access (Admin, Developer, User).
- **Ticket Management System**
  Users should be able to raise tickets, edit details, and track ticket status.
- **Assignment & Reassignment Module**
  Admin should be able to assign or reassign tickets to developers with proper tracking.
- **Task Management**
  Developers should be able to create and manage tasks related to tickets.
- **Status Tracking System**
  Tickets should follow structured workflow such as Open, In Progress, Resolved, and Closed.
- **Comment & Notification System**
  Users and developers should be able to communicate through comments with notification updates.
- **Admin Dashboard**
  Admin panel should provide monitoring of tickets, user activities, and overall system workflow.
- **Secure and Scalable Architecture**
  The system should use proper session handling, input validation, and database security practices.

4.4 SYSTEM FEASIBILITY

4.4.1 Does the System Contribute to Overall Objectives of Organization?

Yes

4.4.2 Can the System Be Implemented Using the Current Technology Within the Given Cost?

Yes

4.4.3 Can the System Be Integrated with Other Systems Which Are Already in Place?

Yes

4.5 ACTIVITY IN NEW SYSTEM

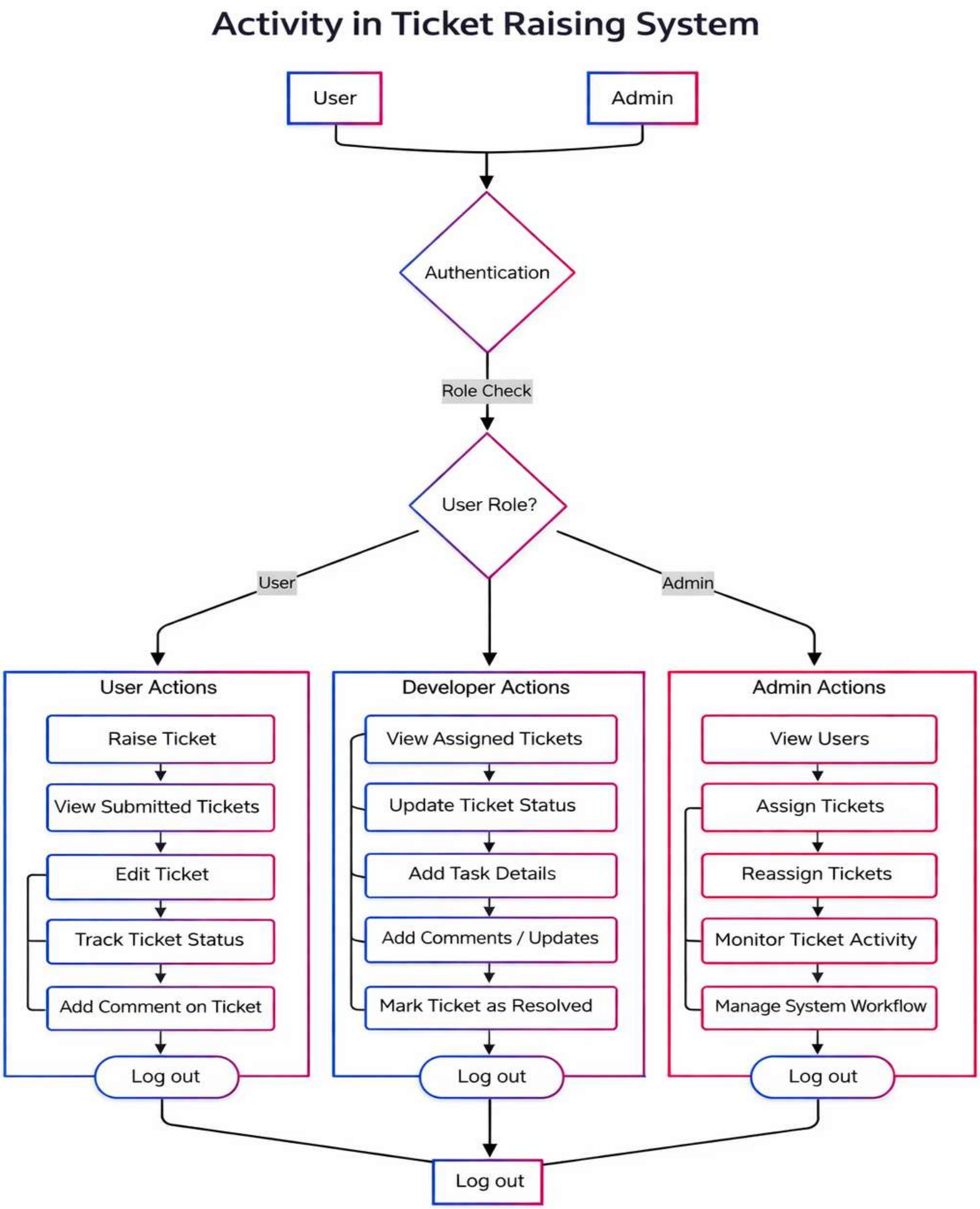

Fig 4.1 Activity in New System

4.6 FEATURES OF NEW SYSTEM

- **Login and Authentication**
  Secure login system for Admin, Developer, and User using session-based authentication. Role-based access control ensures that each user can only access permitted functionalities.

- **Ticket Creation and Management**
  Users can raise new tickets by providing issue details. They can view submitted tickets, edit allowed information, and track ticket status.

- **Ticket Assignment and Reassignment**
  Admin can assign tickets to developers and reassign them when required. Reassignment includes proper tracking for workflow transparency.

- **Task Management System**
  Developers can add task details related to assigned tickets, update progress, and manage resolution activities.

- **Status Tracking Workflow**
  Tickets follow a structured workflow such as Open, In Progress, Resolved, and Closed, allowing systematic issue tracking.

- **Comment and Notification System**
  Users and developers can communicate through comments on tickets. AJAX-based updates allow real-time interaction without page reload.

- **Admin Monitoring and Control**
  Admin dashboard allows monitoring of all tickets, users, assignments, and system activity to ensure proper workflow management.

- **Responsive User Interface**
  The system is developed using Bootstrap to ensure smooth performance and compatibility across different devices.

- **Secure Data Handling**
  Input validation, session management, and database-level security practices are implemented to maintain data integrity and prevent unauthorized access.

4.7 LIST OF MAIN MODULES

The main modules involved in the Ticket Raising System application are:

• **Authentication Module:**
Handles user registration, login, logout, session management, and role-based access control for Admin, Developer, and User.

• **Ticket Management Module:**
Allows users to raise tickets, edit details (if permitted), view submitted tickets, and track ticket status.

• **Assignment & Reassignment Module:**
Enables Admin to assign tickets to developers and reassign them when required with proper tracking.

• **Task Management Module:**
Allows developers to create and update task details related to assigned tickets.

• **Status Management Module:**
Manages ticket workflow such as Open, In Progress, Resolved, and Closed.

• **Comment & Notification Module:**
Enables communication between users and developers through comments. AJAX-based updates allow dynamic notification handling.

• **Admin Dashboard Module:**
Provides complete monitoring of users, tickets, assignments, and overall system workflow.

4.8 SELECTION OF HARDWARE AND SOFTWARE

Hardware Specification:

- A computer or smartphone with at least 4GB RAM for smooth performance.
- Reliable internet connection for accessing the web-based system.
- Android or iOS phone for vendor/customer convenience.

Software Specification:

- **Frontend**: TML, CSS, Bootstrap, JavaScript, jQuery.
- **Backend**: PHP with CodeIgniter Framework.
- **Database**: MySQL.
- **Tools**: VS Code for development,Git for version control and Postman for API testing

CHAPTER 5: - SYSTEM DESIGN

5.1 System Design & Methodology

The system design and methodology for the **Ticket Raising System (TRS)** were structured to ensure modularity, maintainability, and proper role-based workflow management. The system connects three types of users — **Admin, Developer, and User** — and provides structured functionalities such as ticket creation, assignment, status tracking, task management, and comment communication.

The system follows the **MVC (Model-View-Controller) architecture** using the CodeIgniter framework to maintain clear separation between business logic, user interface, and database operations.

System Design

• User Interface

The user interface is developed using **HTML, CSS, Bootstrap, JavaScript, and jQuery**, ensuring a responsive and user-friendly design.

It includes pages such as:

- Login Page
- User Dashboard
- Developer Dashboard
- Admin Dashboard
- Ticket Creation Form
- Ticket Details Page
- Task Management Section

Each user role sees a customized interface based on their permissions.

• Notification System

The system includes an AJAX-based comment and update mechanism that dynamically reflects ticket updates without page reload. Users are notified through system alerts when ticket status changes or comments are added.

• **Role-Based Access Control**

Role-based authentication ensures:

- Users can raise and track tickets
- Developers can update status and manage tasks
- Admin can assign/reassign tickets and monitor workflow

Access restrictions are implemented at controller and session level.

Methodology:

- **Requirement analysis**: he first step involved gathering and analyzing system requirements such as login system, ticket workflow, assignment process, status tracking (Open, In Progress, Resolved, Closed), and comment communication system.
- The goal was to replace manual ticket tracking with a structured and centralized digital system.
- **System architecture design**: **The platform follows a structured MVC architecture:**
- **Frontend:** HTML, CSS, Bootstrap, JavaScript, jQuery
- **Backend:** PHP with CodeIgniter Framework
- **Database:** MySQL
- **User interface design**: The UI was designed using Bootstrap to ensure responsiveness across devices. Emphasis was placed on clarity, simple navigation, and dashboard-based workflow to improve usability.

Forms include proper validation and structured layouts for better user experience.

- **Development**: Development was carried out using:
- VS Code (Code Editor)
- XAMPP (Local Server Environment)
- phpMyAdmin (Database Management)
- GitHub (Version Control)

- Backend controllers were structured according to CodeIgniter MVC principles. AJAX was implemented for dynamic ticket updates without page reload.

- **Testing**: Manual testing was performed to verify:
- Login and session handling
- Role-based access restrictions
- Ticket CRUD operations
- Assignment and reassignment
- Status updates
- Comment functionality
- Input validation and session security were checked to ensure system reliability.

- **Deployment (Future Stage)**: The system was executed on a local server using XAMPP (Apache + MySQL).
- It can be deployed on any hosting server supporting PHP and MySQL for production use.

The system design and methodology of the Ticket Raising System focus on delivering a structured, secure, and role-based issue management platform.
Built using PHP (CodeIgniter) and MySQL, it ensures maintainability, scalability, and efficient workflow management within the organization.

CHAPTER 6: IMPLEMENTATION

6.1 IMPLEMENTATION ENVIRONMENT

The implementation environment for the **Ticket Raising System (TRS)** includes various tools and technologies used for backend development, frontend integration, database management, testing, and deployment.

- **Backend Technology:** The backend is developed using **PHP with CodeIgniter Framework**, which follows the MVC (Model-View-Controller) architecture. It handles business logic, authentication, session management, ticket operations, and database interaction.

- **Frontend Technology:** The user interface is developed using **HTML, CSS, Bootstrap, JavaScript, and jQuery**. AJAX is used to perform asynchronous operations such as ticket updates, comments, and status changes without page reload.

- **Database:** **MySQL** is used as the relational database management system. It stores structured data including users, tickets, tasks, comments, and messages.

- **Local Server Environment:** The application is executed using **XAMPP (Apache + MySQL)** during development.

- **IDE (Development Tool):** The system is developed using **Visual Studio Code (VS Code)** with extensions for PHP, JavaScript, and Git integration.

- **Operating System:** The project can be developed and deployed on major operating systems such as Windows, macOS, or Linux.

- **Testing Tools:** Manual testing was performed for UI and workflow validation. Database testing and management were done using **phpMyAdmin**. Postman was optionally used for route testing if required.

- **Version Control:** **Git and GitHub** were used for version control and maintaining source code backups.

- **Web Browser:** Google Chrome and Mozilla Firefox were used for system testing and debugging.

6.2 MODULE SPECIFICATION

The Ticket Raising System Website is composed of multiple modules, each responsible for a specific functionality:

• User Authentication Module

Enables users to register, login, and logout securely. Session-based authentication is implemented using CodeIgniter sessions. Role-based access control restricts access based on Admin, Developer, and User roles.

• Ticket Management Module

Allows users to:

- Create new tickets
- View submitted tickets
- Edit ticket details (if permitted)
- Track ticket status

This module handles all ticket-related CRUD operations.

• Assignment & Reassignment Module

Allows Admin to:

- Assign tickets to developers
- Reassign tickets when required
- Maintain proper tracking of assignments

This ensures clear responsibility allocation.

• Task Management Module

Developers can:

- Add task details related to assigned tickets
- Update task progress
- Manage issue resolution activities

• Status Management Module

Manages structured workflow including:

- Open
- In Progress
- Resolved
- Closed

Status updates are reflected dynamically using AJAX.

• Comment & Notification Module

Allows communication between users and developers through ticket comments. AJAX-based updates ensure comments appear without page reload. System alerts notify users of status changes and updates.

• **Admin Dashboard Module**

Provides complete system monitoring, including:

- View all tickets
- View user details
- Manage assignments
- Monitor workflow progress

• **Profile Management Module**

Allows users to update their personal information securely, including name, email, and password.

• **Logout Module**

Ends the session securely by destroying session data and redirecting the user to the login page.

6.3 OUTCOMES

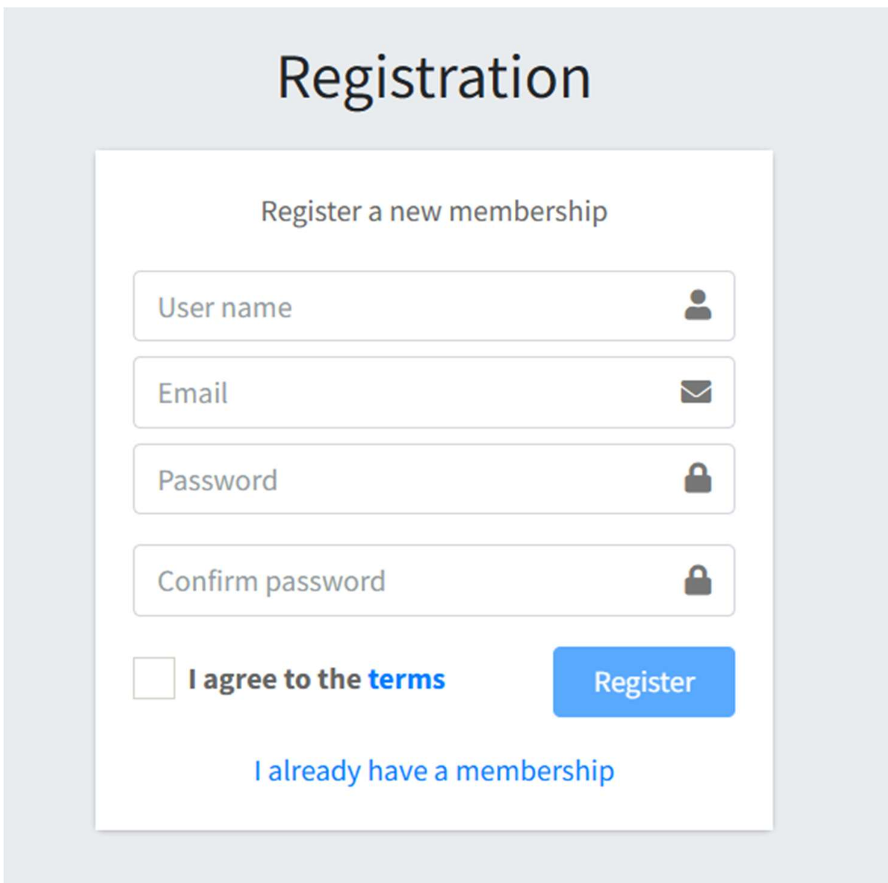

Fig 6.1 Signup Screen

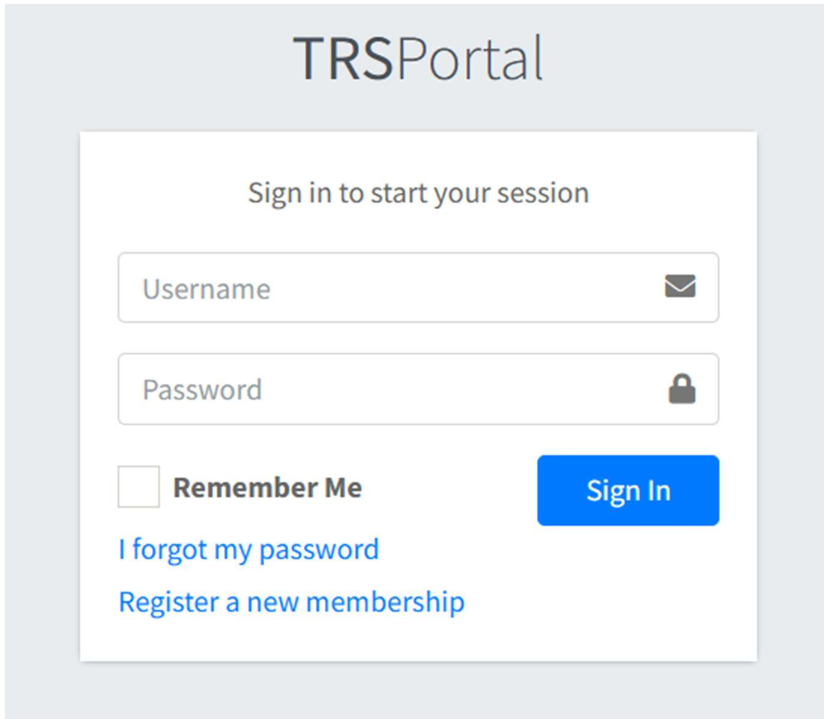

Fig 6.2 Login Screen

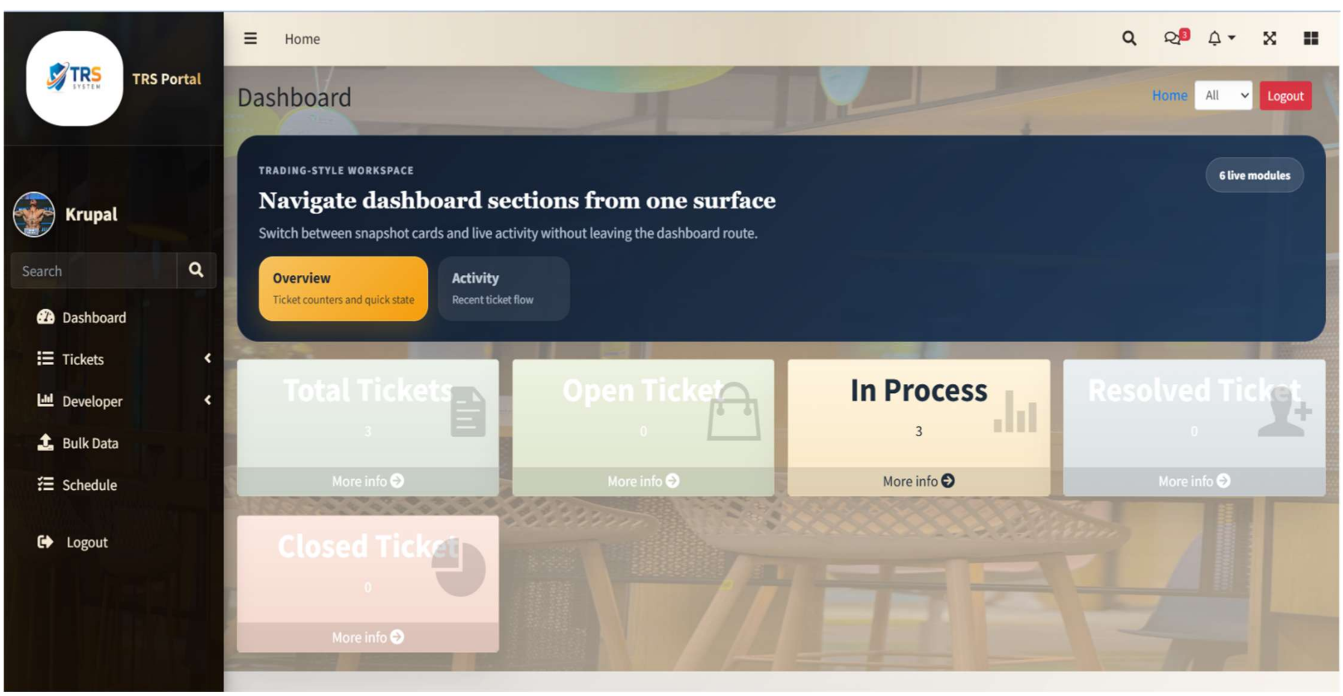

Fig 6.3 Admin Home Page I

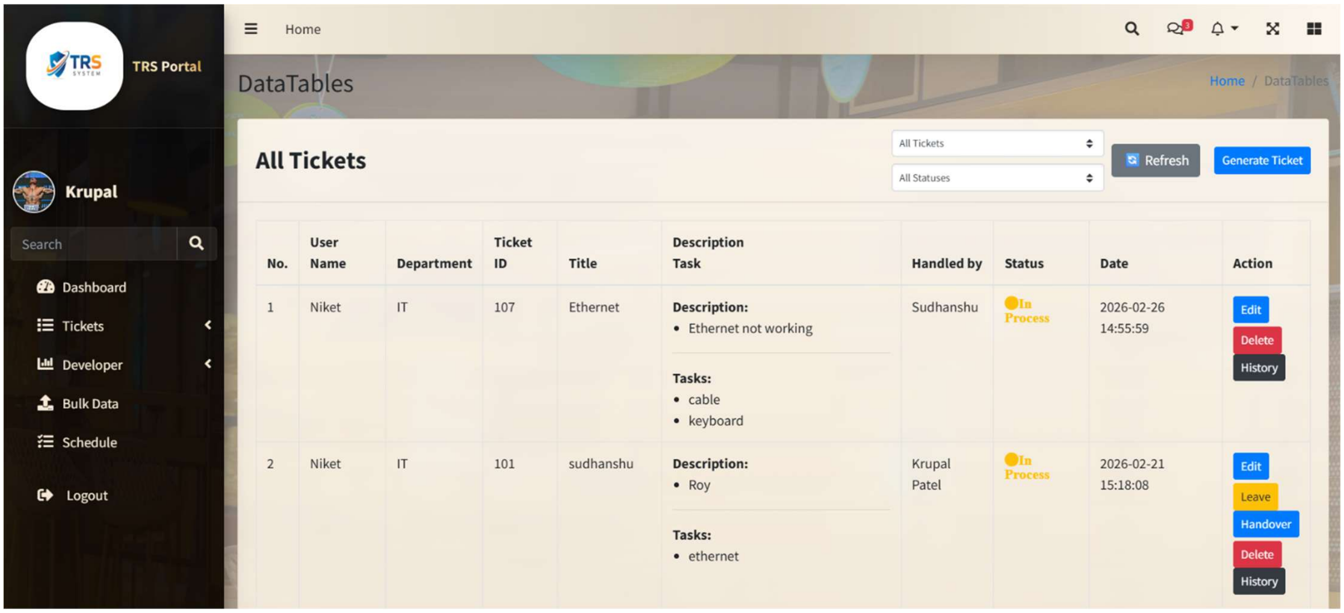

Fig 6.4 Admin List Page

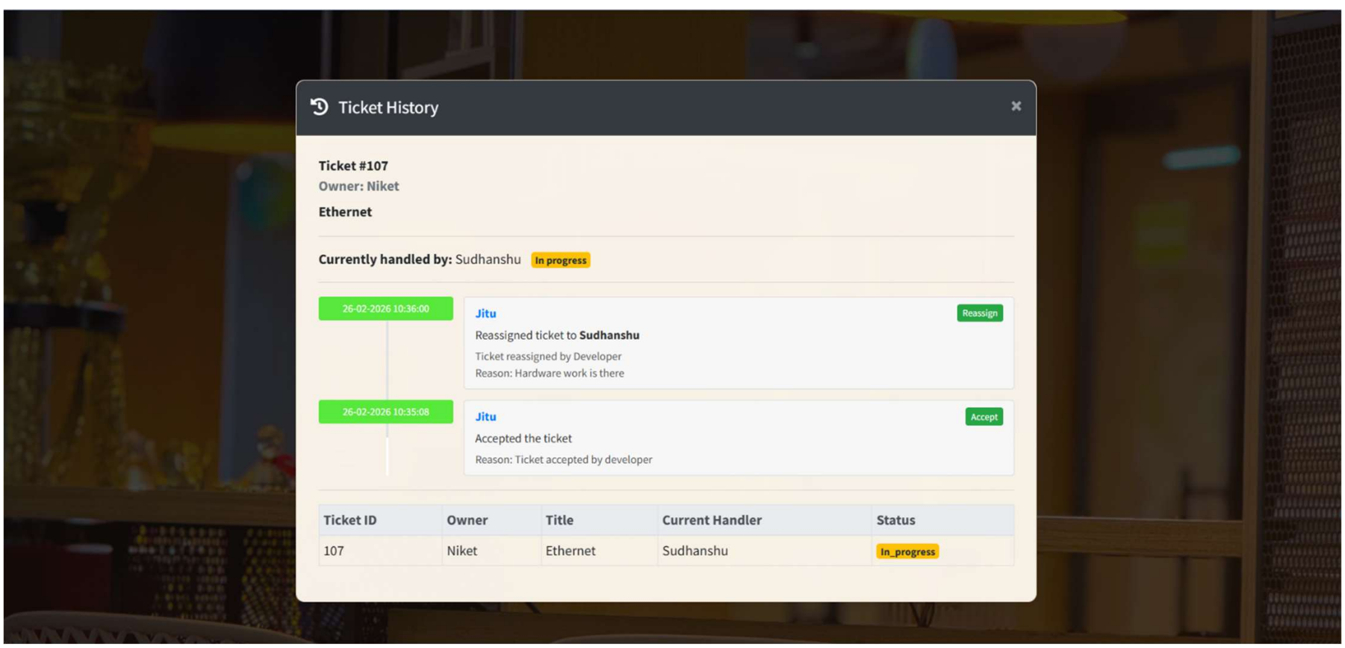

Fig 6.5 Admin/Developer History Page I

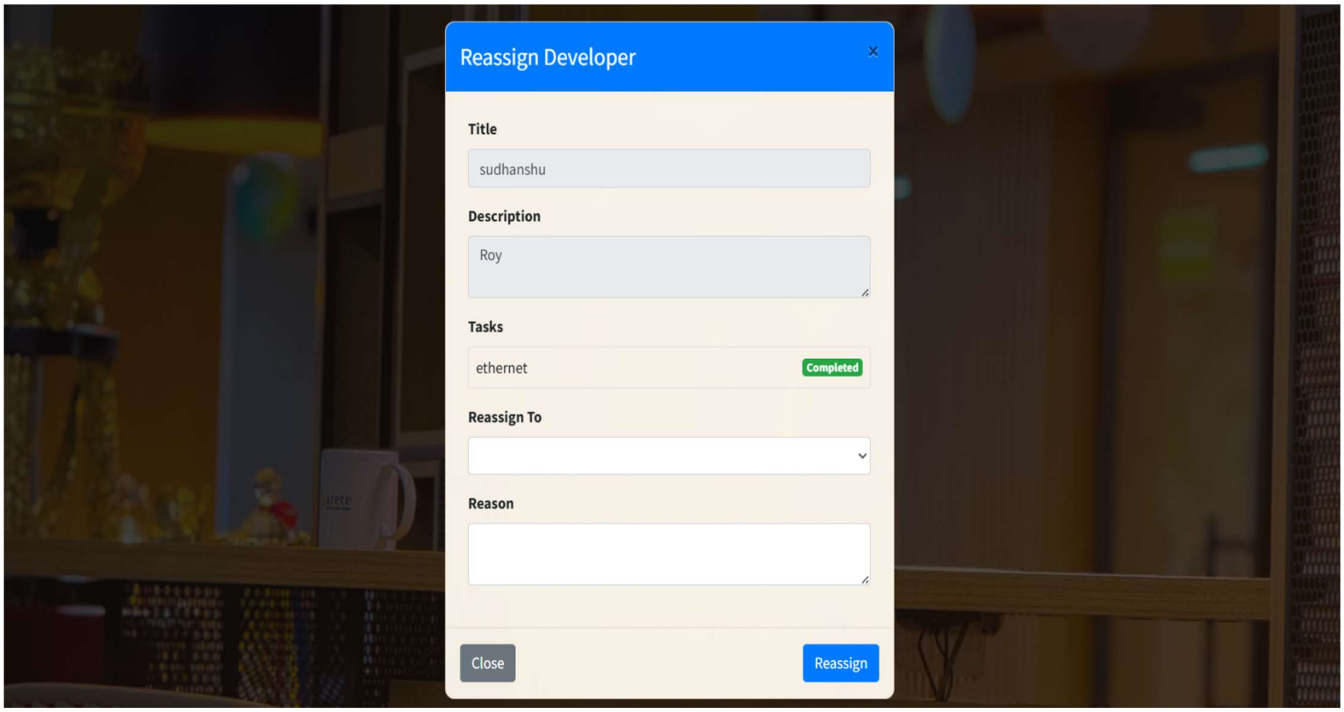

Fig 6.6 Admin/Developer:Handover to other Developer Page

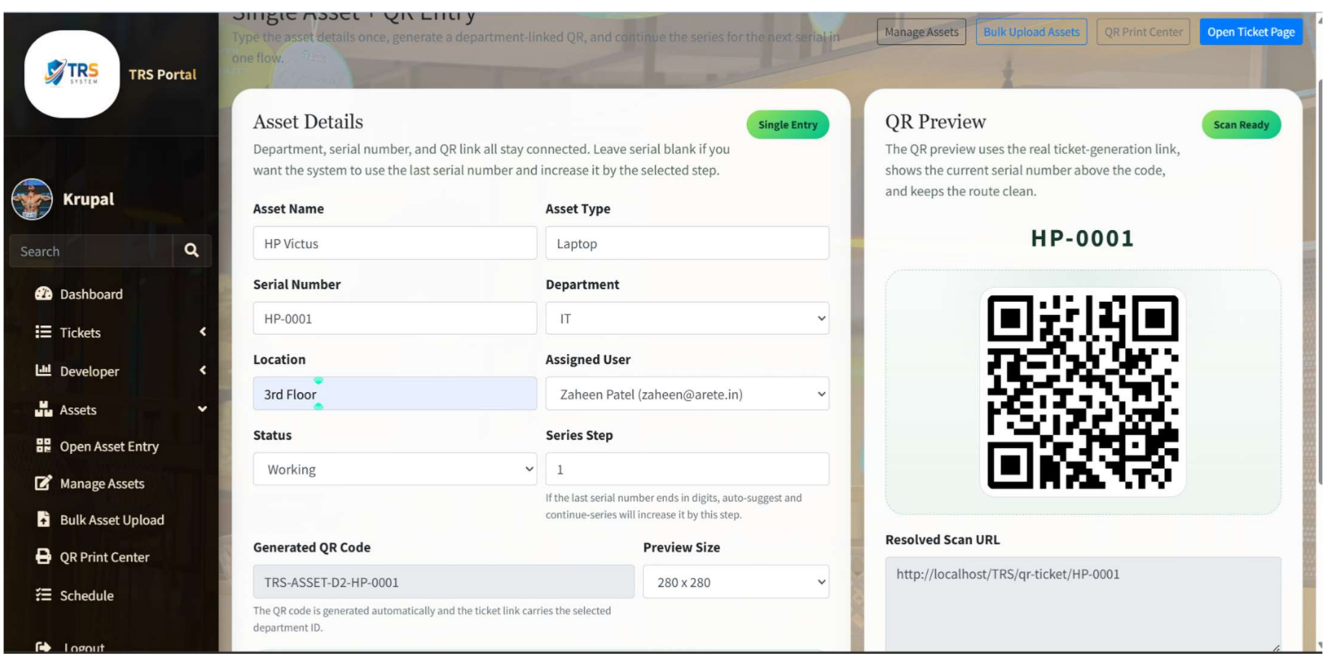

Fig: Admin/Developer:QR Page for manage ticket

Fig 6.7 Admin Developer Status Page

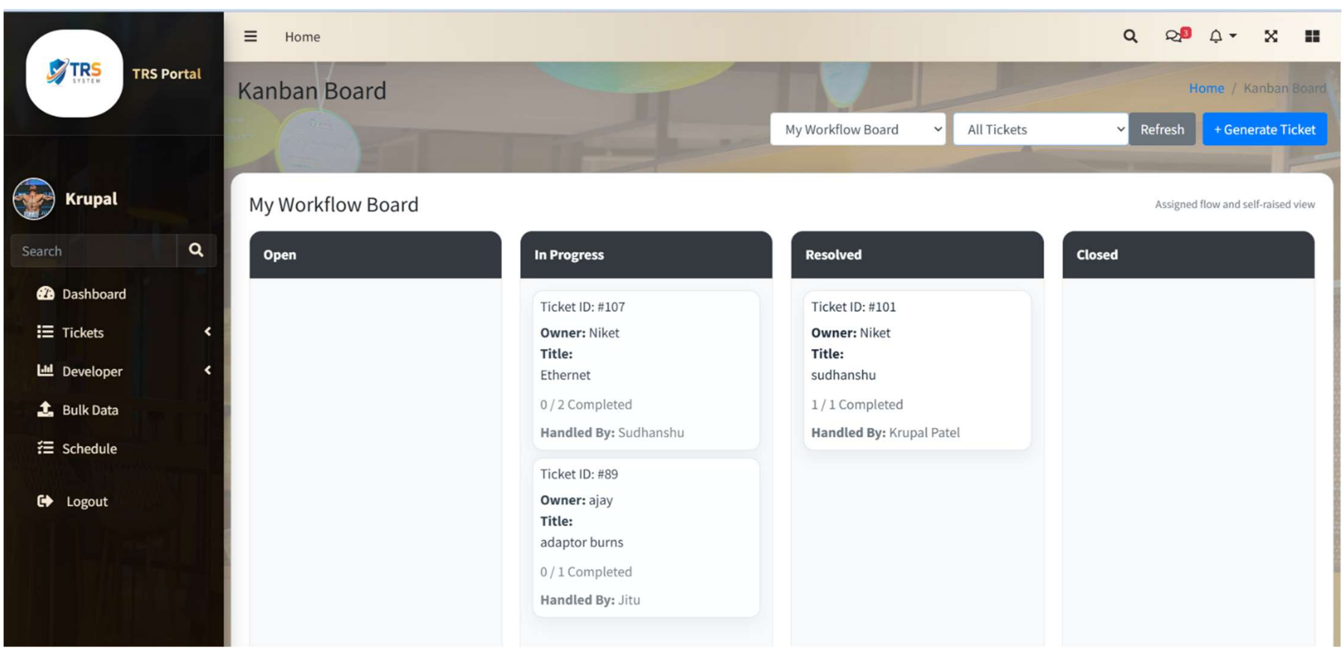

Fig 6.8 Admin Karban Board Page

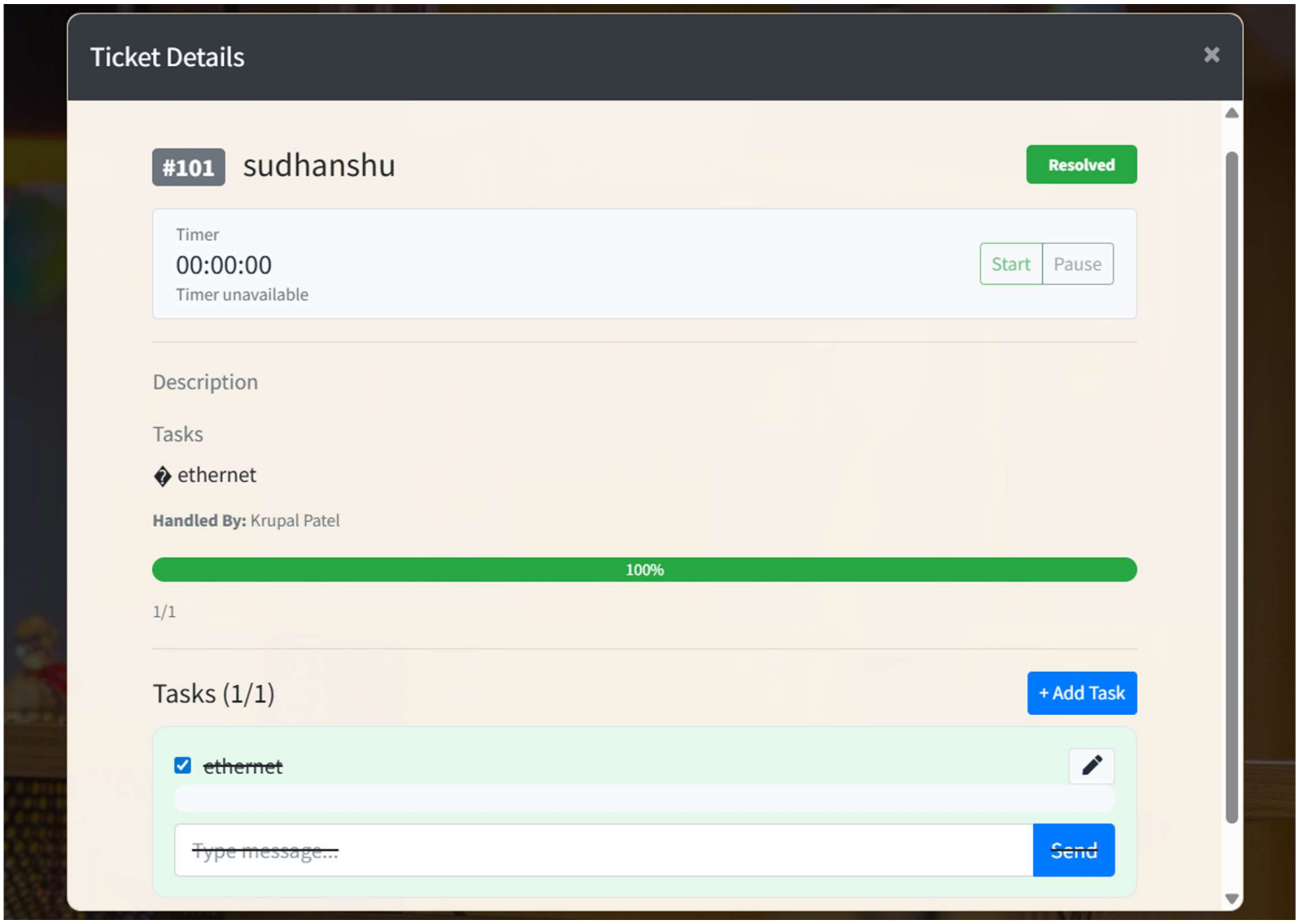

Fig 6.9 Admin:Karban Ticket Detail Page

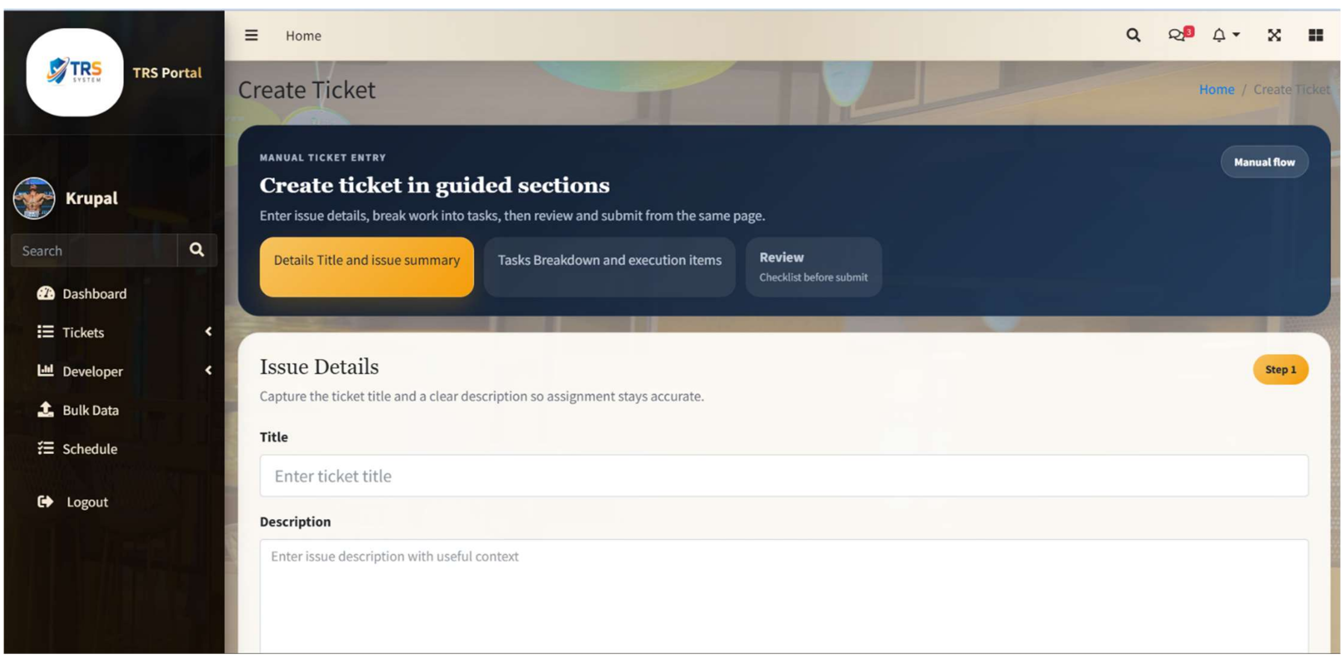

Fig 6.10 Generate Ticket Page

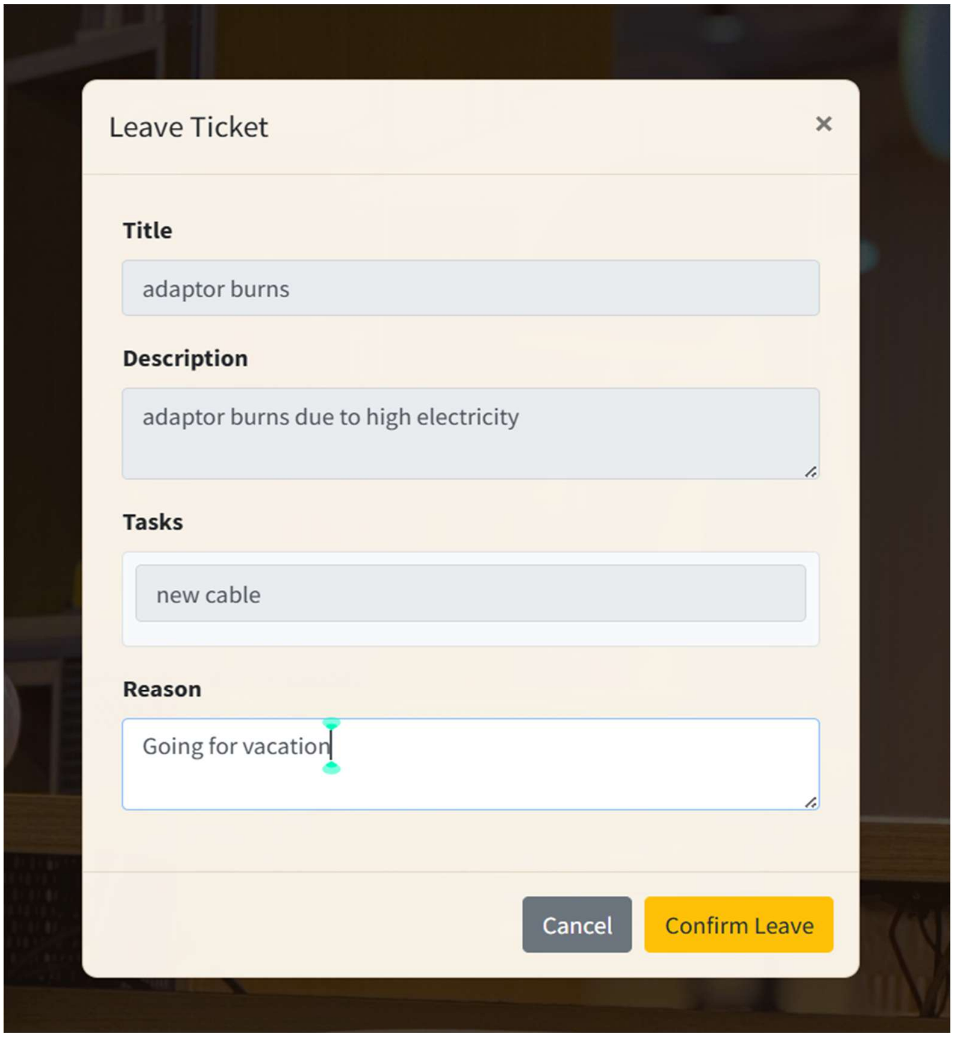

Fig 6.11 Admin/Developer Leave Ticket Page

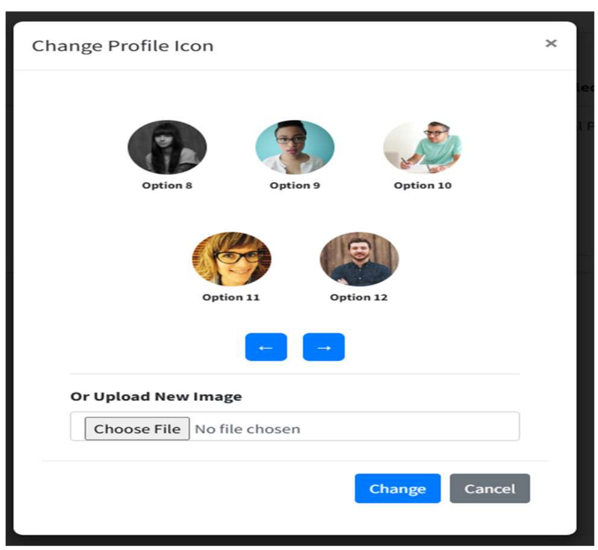

Fig 6.12 Profile Picture Change Page

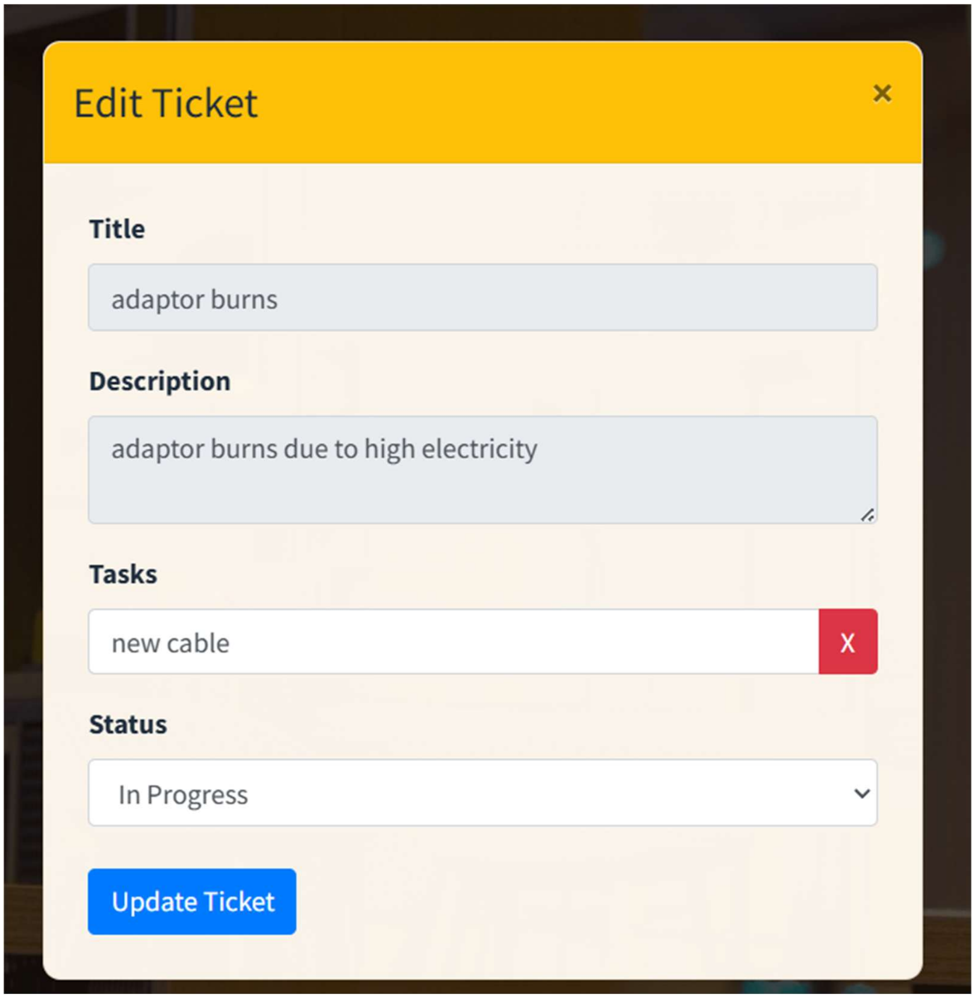

Fig 6.13 Edit Ticket Page

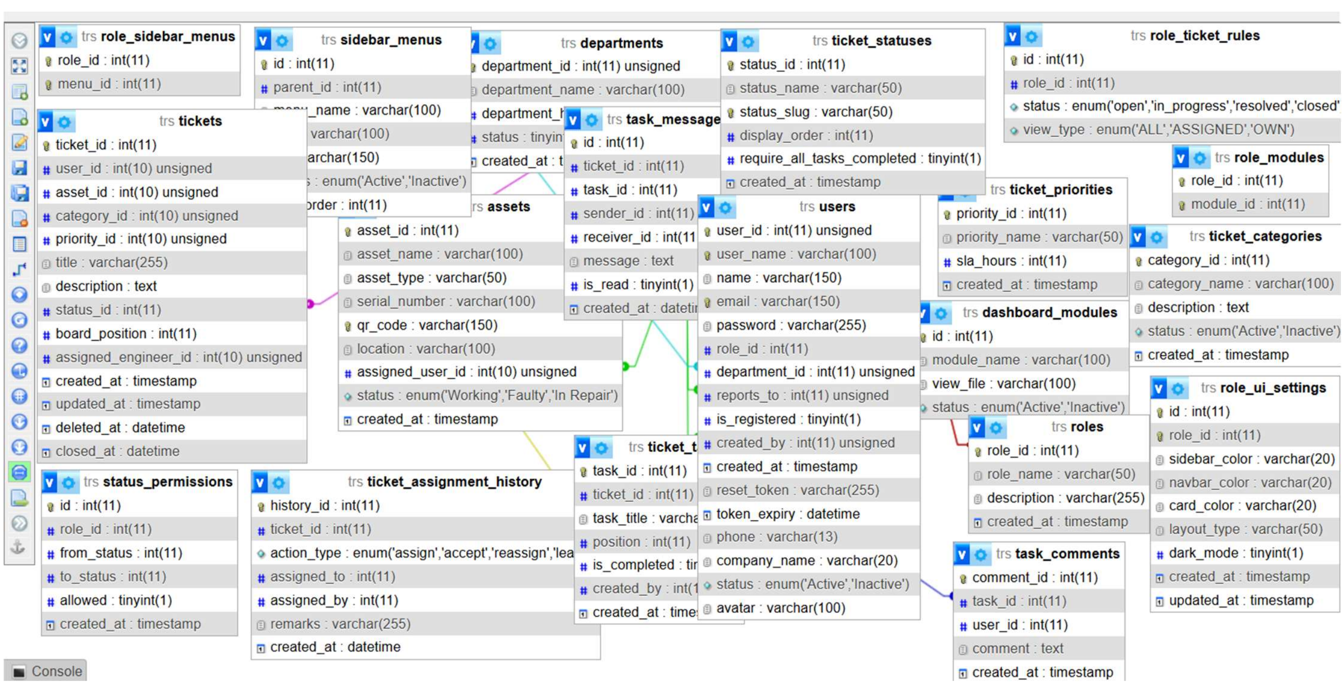

Fig 6.14 Mysql Database Schema Page

6.4 RESULT ANALYSIS

- The Ticket Raising System was successfully implemented according to the defined requirements, and all core modules functioned properly.
- Role-based access, ticket management, assignment, and status updates worked accurately after testing.
- Minor revisions were made during development and were completed effectively within the internship timeline.
- Overall, the system achieved its objective of providing a structured and centralized ticket management solution.

CHAPTER 7: - TESTING

7.1 TESTING PLAN

- **Functional Testing:**
  Ensures that all features of the Ticket Raising System function correctly. It includes testing user authentication, ticket creation, editing and deletion, ticket assignment and reassignment, task management, status updates (Open, In Progress, Resolved, Closed), comment system, and role-based access control for Admin, Developer, and User.
- **Usability Testing:**
  Focuses on user interface and user experience. It involves testing dashboard navigation, form validation, button actions, and overall system usability to ensure smooth interaction for all user roles.
- **Compatibility Testing:**
  Ensures that the Ticket Raising System works properly across different web browsers such as Google Chrome and Mozilla Firefox. It also verifies responsiveness on desktops and different screen sizes using Bootstrap.
- **Performance Testing:**
  Evaluates how the system performs under multiple user interactions, such as handling multiple ticket submissions, status updates, and comment operations simultaneously.
- **Security Testing:**
  Ensures that the system protects user data and restricts unauthorized access. It includes testing session-based authentication, role-based access control, input validation, and protection against SQL injection and other common vulnerabilities.
- **Regression Testing:**
  Ensures that existing functionalities continue to work properly after implementing new features or bug fixes. Core modules such as authentication, ticket workflow, and assignment system are re-tested after updates.

7.2 TEST RESULT AND ANALYSIS

Test ID	Test Condition	Expected Output	Actual Output	Remark
1	Valid login credentials provided	User should be logged in	User successfully logged in	Pass
2	Invalid username provided	Error message should be displayed	Error message displayed	Pass
3	Invalid password provided	Error message should be displayed	Error message displayed	Pass
4	Empty username and password fields	Validation error should be shown	Validation error displayed	Pass
5	Click login without proper input	Input validation message should appear	Validation message displayed	Pass
6	User creates a new ticket	Ticket should be saved in database	Ticket saved successfully	Pass
7	User edits ticket details	Updated information should be reflected	Ticket updated successfully	Pass
8	Admin assigns ticket to developer	Ticket should be assigned successfully	Ticket assigned successfully	Pass
9	Admin reassigns ticket	Ticket should be reassigned with updated developer	Ticket reassigned successfully	Pass
10	Developer updates ticket status	Status should change (Open → In Progress)	Status updated correctly	Pass
11	Developer marks ticket as Resolved	Ticket status should change to Resolved	Status changed to Resolved	Pass
12	User adds comment on ticket	Comment should be saved and displayed	Comment displayed successfully	Pass
13	Developer adds task to ticket	Task details should be stored and visible	Task added successfully	Pass
14	Unauthorized user tries to access admin panel	Access should be restricted	Access denied	Pass
15	Click Logout button	User should be logged out and redirected	User logged out successfully	Pass

Test ID	Test Condition	Expected Output	Actual Output	Remark
16	Invalid ticket ID accessed manually	Error or restricted message should appear	Access restricted	Pass
17	Multiple tickets created	All tickets should display in dashboard	Tickets displayed correctly	Pass
18	AJAX status update without reload	Status should update dynamically	Status updated without page reload	Pass
19	Session timeout	User should be redirected to login page	Redirected to login page	Pass
20	Database record verification	Ticket data should match stored values	Data stored correctly	Pass

Table 7.1 Test Results and Analysis

CHAPTER 8: CONCLUSION AND DISCUSSION

8.1 OVERALL ANALYSIS OF INTERNSHIP

Based on the objectives and scope of the internship at **Arete Services Private Limited**, it provided a valuable opportunity to gain practical experience in full-stack web development using PHP and related technologies. During the internship, I developed a **Ticket Raising System** that streamlined issue tracking and management within an organization.

The internship helped me strengthen my understanding of backend development using PHP (CodeIgniter), database management using MySQL, and frontend integration using HTML, CSS, Bootstrap, JavaScript, and AJAX. I gained real-world experience in implementing role-based access control, session management, ticket workflow, task handling, and dynamic updates without page reload.

Additionally, I improved my debugging skills, database design knowledge, and problem-solving abilities while working on a live project environment. Tools such as VS Code, XAMPP, phpMyAdmin, GitHub, and Postman enhanced my development workflow.

Overall, the internship was a structured and practical learning experience that bridged the gap between academic knowledge and real-world application. It provided a strong foundation for future growth in web development and software engineering.

8.2 DATE OF CONTINUOUS EVALUATION (CE-I AND CE-II)

- **CE-I:** 25-02-2026
- **CE-II:** 25-03-2026

8.3 PROBLEM ENCOUNTERED AND POSSIBLE SOLUTIONS

- Since the project was built using PHP (CodeIgniter) and MySQL, understanding MVC architecture and proper controller-model interaction was initially challenging. I overcame this by studying official documentation and practicing small modules step-by-step.

- Implementing role-based access control and session management required careful logic handling. I resolved issues by debugging session variables and testing user roles thoroughly.

- Handling AJAX-based updates without page reload caused some integration challenges between frontend and backend. These issues were solved by carefully checking response formats and validating JSON outputs.

- Designing relational database tables and maintaining proper relationships between users, tickets, tasks, and comments required optimization. I improved this by refining database schema and validating foreign key relationships.

- Managing reassignment logic and ticket status workflow required multiple revisions. Breaking the functionality into smaller modules helped in debugging and improving system stability.

- Implementing modules such as authentication, ticket assignment, task management, and role-based access control required clear logic structuring. Breaking the system into smaller modules and seeking guidance from senior developers helped me overcome these challenges effectively.

- Integrating frontend (HTML, Bootstrap, jQuery) with backend controllers using AJAX initially caused some issues in handling responses and updating data dynamically. Debugging these issues and testing requests carefully helped improve system stability.

- The project required optimizing MySQL database queries and ensuring efficient data retrieval for tickets, tasks, and comments. I learned about proper indexing, query structuring, and maintaining relational integrity to enhance overall performance.

8.4 SUMMARY OF INTERNSHIP

During my internship at Arete Services Private Limited as a MERN Stack Developer from January 20, 2026, to April 19, 2026, I gained extensive technical knowledge and hands-on experience in full-stack development. I worked on the **Ticket Raising System** project, which involved designing and implementing key features such as user authentication, business listings, bookmarks, and reviews. This internship allowed me to refine my skills in **React.js, Node.js, Express.js,**

MongoDB, and Tailwind CSS, while also improving my problem-solving and debugging abilities. Additionally, I enhanced my non-technical skills, including teamwork, communication, and time management. Working in a professional environment provided me with insights into how to handle real-world development challenges, meet project deadlines efficiently, and optimize code for better performance and scalability.

8.5 LIMITATION AND FUTURE ENHANCEMENT

One of the limitations of the project is its **compatibility with older web browsers**, as the platform relies on modern web technologies, which may not function efficiently on outdated browser versions. This can restrict access for users who are unable to upgrade due to system limitations. Additionally, as the platform grows, **scalability challenges** may arise. Handling an increasing number of business listings and user traffic efficiently requires robust database management and performance optimization techniques. Future enhancements should focus on **implementing caching mechanisms, load balancing, and server-side optimizations** to improve scalability. Furthermore, integrating **AI- based recommendations and advanced analytics** for vendors and users could enhance the overall experience, making the platform more dynamic and user-friendly.

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